

GENAI & DATA SCIENCE SPECIALIZATION

9-Day Industry Project Execution & Evaluation Framework

Purpose

The objective of this specialization is not to teach individual tools or models in isolation. The core purpose is to train students to execute a complete, industry-level GenAI & Data Science project using modern, AI-assisted workflows. Students will work on real-world business problems across multiple domains and build end-to-end, production-grade solutions integrating ML, Deep Learning, NLP, Small Language Models (SLM), Low-Level Design (LLD), Retrieval-Augmented Generation (RAG), and Agentic AI to deliver measurable business impact.

At the end of 9 days, students will be able to deliver production-grade data science and generative AI solutions with strong business relevance, technical rigor, and deployment readiness.

3-Level Progression Framework

Students must successfully complete the current level before advancing to the next.

Level 1 Foundation Data Scientist Day 1 – Day 3	Objective: Build strong foundations in ML pipelines, ensemble methods, and deep learning models through hands-on industry use cases. Mandatory: Minimum 80% on Level 1 tasks. Students cannot proceed without completing Level 1. Progress Weightage: 40%
Level 2 Core Data Scientist Day 4 – Day 6	Objective: Master sequence modeling, advanced NLP pipelines, BERT fine-tuning, and efficient Small Language Models for domain-specific tasks. Mandatory: Complete Level 1 + minimum 75% on Level 2 tasks. Without completing Level 2, students cannot proceed to Level 3. Progress Weightage: 35% (Cumulative: 75%)
Level 3 Advanced GenAI Engineer Day 7 – Day 9	Objective: Architect production AI systems, implement Retrieval-Augmented Generation, and build autonomous Agentic AI solutions with real-world deployment. Mandatory: Complete Levels 1 & 2 with required passing scores. Progress Weightage: 25% (Cumulative: 100%)

Student Evaluation Model

Level 1 Evaluation – 40% Weightage		
Criteria	Weight	Description
ML Pipeline & Exploratory Data Analysis	15%	End-to-end supervised learning pipeline with EDA, feature engineering, and model evaluation.
Ensembles & Hyperparameter Tuning	15%	Ensemble modeling (RF, XGBoost, LightGBM), tuning strategies, and Streamlit deployment.
Deep Learning (MLP + CNNs)	10%	Neural network design, CNN architecture, transfer learning, and FastAPI deployment.
Passing Score: 32 / 40		

Level 2 Evaluation – 35% Weightage		
Criteria	Weight	Description
Sequence Models & Demand Forecasting	12%	LSTM, attention mechanisms, and time-series forecasting for supply chain use cases.
NLP Pipeline & BERT Fine-Tuning	13%	Full NLP pipeline with tokenization, NER, sentiment analysis, and BERT fine-tuning.
SLM Fine-Tuning, Quantization & Deployment	10%	Fine-tuning small language models, quantization (GGUF), and edge deployment via Ollama.
Passing Score: 28 / 35		

Level 3 Evaluation – 25% Weightage		
Criteria	Weight	Description
Low-Level Design (LLD) for AI Systems	8%	System architecture diagrams, API contracts, data schemas, and production design docs.
RAG, Vector Databases & Prompt Engineering	9%	End-to-end RAG pipeline with vector DB integration, embedding models, and prompt design.
Agentic AI & System Integration	8%	Autonomous agent design, MCP integration, tool use, and final system demo.
Passing Score: 20 / 25		

9-Day Execution Plan

Day	Module & Domain	Domain	Topics Covered	Deliverables	Focus
Day 1	ML – Supervised Learning Churn Prediction & Classification Pipeline	FinTech / Insurance	Supervised learning, EDA, feature engineering, model selection, evaluation metrics	Complete ML Pipeline Notebook, Model Evaluation Report	Level 1 Focus
Day 2	ML – Ensembles & Hyperparameter Tuning Healthcare Premium Prediction	Healthcare / Hospital Network	Ensemble methods (RF, XGBoost, LightGBM), hyperparameter tuning, Streamlit app	Tuned XGBoost Model + Streamlit App	Level 1 Focus
Day 3	DL – Neural Networks & CNNs Deep Learning from Scratch to Transfer Learning	Automotive / Insurance	MLP, CNN architectures, transfer learning, FastAPI deployment	CNN Model with FastAPI + Streamlit Deployment	Level 1 Completion Review
Day 4	DL – LSTM, Attention & Sequence Models Demand Forecasting	Supply Chain / FMCG	LSTM, attention mechanisms, sequence modeling, time-series forecasting	LSTM + Attention Forecasting Solution	Level 2 Focus
Day 5	NLP – Full NLP Pipeline Complaint Classification with BERT Fine-Tuning	Customer Support / Telecom	Tokenization, NER, sentiment analysis, BERT fine-tuning, stage-wise comparison	BERT Model + Stage-wise Comparison Report	Level 2 Focus
Day 6	SLM – Small Language Models Fine-Tuning, Quantisation & Edge Deployment	Legal Tech / Healthcare Documentation	SLM fine-tuning, quantization (GGUF), edge deployment with Ollama	Fine-tuned & Quantized SLM with Ollama Deployment	Level 2 Completion Review
Day 7	LLD – Low-Level Design Architecting AI/ML Systems for Production	E-Commerce / Platform Engineering	System architecture, API contracts, data schemas, LLD diagrams for AI pipelines	Complete LLD Document (Diagrams + API Contracts)	Level 3 Focus
Day 8	Gen AI – RAG, Vector DBs & Prompt Engineering Real Estate Assistant	Real Estate / PropTech	RAG pipeline, vector databases, embedding models, prompt engineering, Streamlit UI	Fully Functional RAG Application with Streamlit UI	Level 3 Focus
Day 9	Gen AI – Agentic AI & MCP Autonomous HR Onboarding Agent	HR Tech / Enterprise SaaS	Agentic AI, MCP, autonomous planning, tool use, system integration, final demo	Autonomous Agent, Final Report, LLD Update, PPT Presentation & Demo	Level 3 Completion Review

Progress Monitoring Matrix

Day	Expected Progress	Cumulative Milestone Target
Day 1	12%	ML Pipeline & Supervised Learning Baseline Established
Day 2	25%	Ensemble Models Tuned & Streamlit App Deployed
Day 3	40%	Level 1 Complete (CNN Model Deployed via FastAPI)
Day 4	52%	LSTM Forecasting Solution Validated
Day 5	63%	BERT Fine-Tuned NLP Pipeline Operational
Day 6	75%	Level 2 Complete (SLM Quantized & Edge-Deployed via Ollama)
Day 7	83%	Production LLD Documented & AI System Architecture Finalized
Day 8	92%	RAG Application with Vector DB Fully Functional
Day 9	100%	Level 3 Complete (Agentic AI System Delivered & Final Pitch Given)

Daily Evaluation Plan

Each day is assessed against Key Learning Outcomes (KLOs) tied to the day's module. The evaluation methods below apply per day and feed into the cumulative level scores above.

Day 1 — ML: Supervised Learning & Churn Prediction

Key Learning Outcomes

✓ Formulate a supervised ML problem from a business brief (churn prediction).
✓ Execute an end-to-end ML pipeline: EDA, preprocessing, feature engineering, model training.
✓ Evaluate model performance using accuracy, F1, ROC-AUC and interpret results.
✓ Justify model selection and articulate limitations and potential biases.

Evaluation Methods

- Practical Exercise: Build a complete classification pipeline on the FinTech churn dataset.
- Model Evaluation Report: Document preprocessing decisions, metric analysis, and business insights.
- Discussion: Justify algorithm choices and explain impact on business outcomes.

Day 2 — ML: Ensembles & Hyperparameter Tuning

Key Learning Outcomes

✓ Compare ensemble methods: Random Forest, XGBoost, LightGBM, and Gradient Boosting.
✓ Apply hyperparameter tuning strategies (GridSearch, RandomSearch, Optuna).
✓ Deploy a tuned model via a Streamlit interface for stakeholder interaction.
✓ Interpret feature importance and business impact of predictions.

Evaluation Methods

- Practical Exercise: Train and tune XGBoost on healthcare premium data.
- Streamlit App: Deploy interactive prediction interface.
- Discussion: Compare ensemble strategies and tuning trade-offs.

Day 3 — Deep Learning: CNNs & Transfer Learning

Key Learning Outcomes

✓ Differentiate between MLP, CNN, and Transfer Learning architectures.
✓ Understand backpropagation, optimizers (Adam, SGD), and regularization techniques.
✓ Apply transfer learning using pre-trained models (ResNet, EfficientNet).
✓ Deploy a trained CNN via FastAPI and wrap it with a Streamlit UI.

Evaluation Methods

- Practical Exercise: Build and train a CNN for the automotive/insurance image classification task.
- Deployment: FastAPI endpoint + Streamlit UI operational.
- Level 1 Completion Review: Combined evaluation across Days 1–3 (passing score: 32/40).

Day 4 — Deep Learning: LSTM & Sequence Modeling

Key Learning Outcomes

✓ Explain RNNs, LSTMs, and attention mechanisms and when to use each.
✓ Build an LSTM model for time-series demand forecasting.
✓ Incorporate attention layers to improve sequence model performance.
✓ Evaluate forecasting accuracy using RMSE, MAE, and MAPE.

Evaluation Methods

- Practical Exercise: LSTM + Attention forecasting on supply chain demand data.
- Evaluation: Metric comparison across baseline, LSTM, and attention-augmented models.
- Discussion: Trade-offs between LSTM vs Transformer-based sequence models.

Day 5 — NLP: Full Pipeline & BERT Fine-Tuning

Key Learning Outcomes

✓ Apply text preprocessing: tokenization, stemming, lemmatization, stop-word removal.
✓ Implement NLP tasks: sentiment analysis, NER, and multi-class text classification.
✓ Fine-tune a BERT model on a domain-specific complaint classification dataset.
✓ Compare TF-IDF, word embeddings, and transformer-based representations.

Evaluation Methods

- Practical Exercise: End-to-end complaint classification pipeline with BERT.
- Stage-wise Comparison Report: Baseline vs fine-tuned BERT performance analysis.
- Discussion: Contextual representation benefits and trade-offs vs lighter NLP approaches.

Day 6 — Small Language Models: Fine-Tuning & Edge Deployment

Key Learning Outcomes

✓ Articulate SLM advantages (efficiency, latency, cost) over large LLMs.
✓ Fine-tune an SLM for a domain-specific documentation task (legal/healthcare).
✓ Quantize a fine-tuned model to GGUF format for reduced footprint.
✓ Deploy the quantized SLM locally using Ollama for edge inference.

Evaluation Methods

- Practical Exercise: Fine-tune, quantize, and deploy SLM on domain documentation.
- Deployment Validation: Ollama-served model with inference latency benchmarks.
- Level 2 Completion Review: Combined evaluation across Days 4–6 (passing score: 28/35).

Day 7 — Low-Level Design for AI/ML Systems

Key Learning Outcomes

✓ Understand the role of LLD in building robust, maintainable AI systems.
✓ Design system architecture diagrams for ML pipelines and AI microservices.
✓ Define API contracts (REST/gRPC), data schemas, and inter-component data flow.
✓ Articulate non-functional requirements: scalability, security, latency, observability.

Evaluation Methods

- Design Exercise: Produce a complete LLD document for an e-commerce AI platform.
- Deliverables: Architecture diagrams, API contracts, data models, and sequence diagrams.
- Discussion: Justify design decisions for scalability and production deployment.

Day 8 — Generative AI: RAG, Vector Databases & Prompt Engineering

Key Learning Outcomes

✓ Explain the RAG architecture and its advantages over pure LLM generation.
✓ Build an end-to-end RAG pipeline: document ingestion, chunking, embedding, retrieval.
✓ Integrate a vector database (FAISS, Chroma, Pinecone) for semantic search.
✓ Design and optimize prompts for accurate, context-grounded LLM responses.

Evaluation Methods

- Practical Exercise: Build a fully functional real estate Q&A assistant with RAG.
- Streamlit UI: Interactive interface for document upload and query resolution.
- Discussion: Retrieval quality analysis and prompt engineering strategies.

Day 9 — Agentic AI, MCP & Final Capstone

Key Learning Outcomes

✓ Explain agentic AI: autonomous perception, reasoning, planning, and action loops.
✓ Build an autonomous HR onboarding agent using MCP (Model Context Protocol).
✓ Integrate multiple tools and APIs into a coherent agentic workflow.

✓ Recognize safety considerations and human-in-the-loop integration strategies.

Evaluation Methods

- Practical Deliverable: Autonomous HR onboarding agent with tool use and MCP.
 - Final Report + Updated LLD: Document agent architecture and system integration.
 - Capstone Presentation: Live demo, PPT, and Q&A defending design and ethical considerations.
 - Level 3 Completion Review: Combined evaluation across Days 7–9 (passing score: 20/25).
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General Evaluation Rubric

The following rubric applies across all three levels and all 9 days of the specialization.

Criteria	Weight	Description
Technical Execution & Modeling	35%	Quality and correctness of code, models, and pipelines — including training, evaluation, and optimization rigor.
Business Understanding & Impact	25%	Relevance and measurability of the solution to the stated industry problem, translated into clear business value.
Critical Thinking & Experimentation	20%	Model comparison, ablation studies, error analysis, and iterative optimization with justified design choices.
System Design & Deployment	10%	LLD quality, API contracts, deployment readiness, and production-grade system thinking.
Communication & Documentation	10%	Clarity of reports, presentations, notebook structure, and professional delivery of findings.

Grading Scale Thresholds

Grade	Description
Excellent (90–100%)	Production-ready solutions, deep business insights, innovative model optimizations, and flawless deployment.
Good (80–89%)	Strong technical delivery with clear business alignment, solid documentation, and effective deployment.
Satisfactory (70–79%)	Meets minimum requirements with acceptable performance but shows gaps in business value or optimization depth.
Needs Improvement (<70%)	Structural gaps in modeling, deployment failures, insufficient evaluation, or weak business understanding.

Certification Matrix & Badges

Based on cumulative performance scores and milestone tracking, students will be awarded the following professional certifications:

Completion Milestone	Certification Awarded
40% Completion Layer	Certified Foundation Data Scientist
75% Completion Layer	Certified Core Data Scientist

100% Completion Layer	Certified Advanced GenAI Engineer
90%+ Cumulative Track Score	Certified Industry-Ready GenAI & Data Scientist

Success Criteria

A student is considered successful only when all of the following are met:

1. All three progression levels are completed with the required minimum passing scores.
2. Every core deliverable (ML notebooks, ensemble models, CNN deployment, LSTM forecasting, BERT classifier, SLM deployment, LLD document, RAG application, Agentic AI system) is fully submitted and operational.
3. End-to-end solutions demonstrate clear, measurable business value tied to the given industry domain.
4. The final professional capstone presentation and live system demo are delivered confidently and seamlessly.

This framework ensures students gain hands-on, industry-relevant experience across the full GenAI & Data Science stack — from foundational ML pipelines to autonomous AI systems — preparing them for immediate deployment in modern AI-driven organizations.